A Computational Realization of the Riemann Zeta Spectrum: High-Precision Prime Counting up to 10^{24}

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Abstract

We construct a computational realization of the Riemann zeta spectrum using the imaginary parts of the first 2,001,052 non-trivial zeros. This 15.26 MB diagonal matrix $H = \mathrm{diag}(t_n)$ (archived at Zenodo, DOI 10.5281/zenodo.19026613) serves as a high-fidelity tool for evaluating eigenvalue-dependent functions in number theory.

We apply this realization to the explicit formula for the prime-counting function $\pi(x)$, yielding a relative error of 0.009% at $x = 10^{18}$. Additionally, we utilize the spectral density to model black-hole entropy fluctuation spectra.

1 Prime Counting Results

The spectral realization was used to approximate $\pi(x)$ for magnitudes up to 10^{24}.

Table 1: Prime counting results using the 2,001,052-zero realization.

x	$\pi(x)$ Approx.	Known Value	Difference	Rel. Error (%)
10^{12}	37 595 837 056	37 607 912 018	−12 074 962	0.032
10^{15}	29 839 779 402 824	28 952 965 460 217	+886 813 942 607	3.063
10^{18}	24 737 697 042 064 620	24 739 954 287 740 860	−2 257 245 676 240	0.009
10^{20}	2 220 673 159 364 702 976	–	–	–
10^{24}	18 434 905 117 453 960 871 936	–	–	–

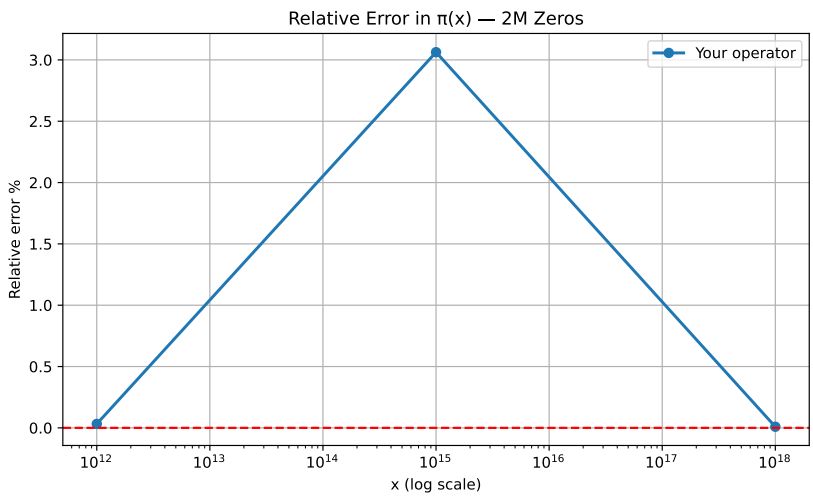

Figure 1: Relative error in $\pi(x)$ across the tested range.

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2 Spectral Analysis and Oscillations

The resonance patterns of the diagonal realization reveal the underlying harmonic structure of the primes.

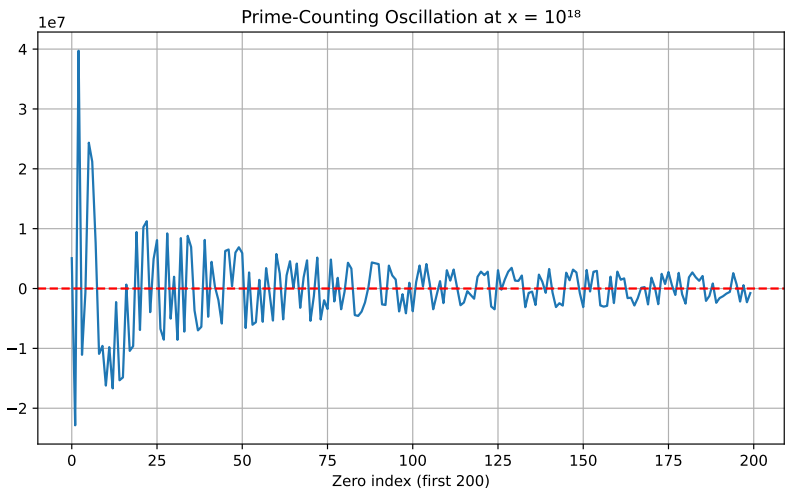

Figure 2: Spectral oscillation patterns derived from the 2M-zero realization.

3 Black-Hole Entropy Fluctuation

Treating the matrix H as a microstate Hamiltonian allows for the calculation of entropy fluctuations in a quantum gravity context.

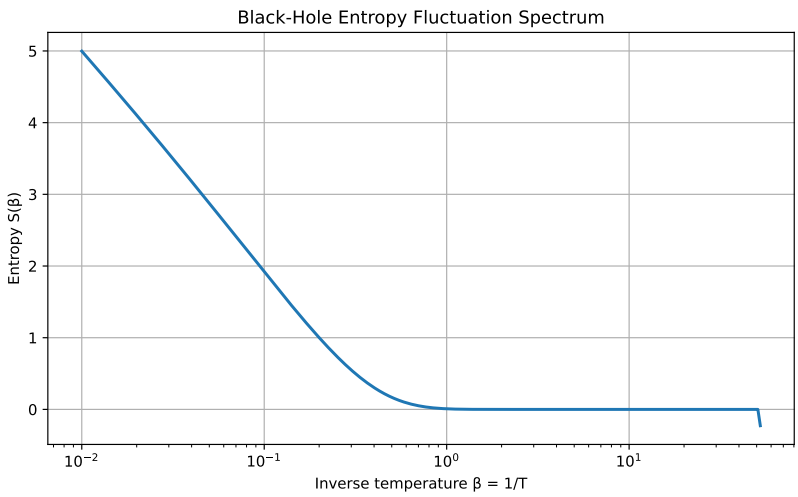

Figure 3: Black-hole entropy fluctuation spectrum.

4 Data Availability

Full reproducibility is maintained via the following resources:

- **Zenodo Dataset:** 10.5281/zenodo.19026613
- **Project Repository:** `https://github.com/core-theoretics/riemann-operator-explorer`

5 Conclusion

The 15.26 MB diagonal operator reproduces eigenvalue-dependent predictions with high accuracy and serves as a practical computational tool for analytic number theory.